

A Quantum Approximate Optimization Algorithm

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1. ABSTRACT & THEORETICAL FOUNDATION

We introduce QAOA, a shallow-depth hybrid quantum algorithm that alternates between problem cost Hamiltonians and transverse mixer Hamiltonians to approximate solutions for NP-hard graph problems like Max-Cut.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"Solving a giant logistics puzzle by vibrating the system back and forth between the rules of the puzzle and a mixing wave until the best solution emerges."

3. Mathematical Formulation & Unitary Rule

$$U(\gamma, \beta) = e^{-i\beta_p H_M} e^{-i\gamma_p H_C} \dots e^{-i\beta_1 H_M} e^{-i\gamma_1 H_C}$$

Trotterized digitization of adiabatic state evolution between driver mixer and problem Hamiltonian.

4. Step-by-Step Circuit Sequence

Initial $H^{\otimes n}$, p blocks of [$R_{zz}(2\gamma)$ cost gates $\rightarrow$ $R_x(2\beta)$ mixer gates], measurement.

5. Executable IBM Qiskit (Python) Code

```
from qiskit import QuantumCircuit
import numpy as np

gamma, beta = np.pi/4, np.pi/3
qc = QuantumCircuit(2)

# Initial superposition
qc.h([0, 1])
```

6. Computational Complexity & Speedup

Classical Time:

Quantum Time:

Speedup Class:

NP-hard (e.g., Max-Cut approximation) Polynomial $O(n^2)$ circuit depth (will be improved) Approximation ratio improves with p

7. Key Applications & Future Research Directions

- Graph Max-Cut and Maximum Independent Set problems
- Traffic light optimization and autonomous vehicle routing
- Financial index tracking and supply chain scheduling

Future Work:

Warm-starting strategies, parameter transferability across graph instances, and constraint-preserving mixers.